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Dynamic wheeled motion control of wheel-biped transformable robots

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ABSTRACT

Most existing biped robots can only walk with their feet or move by wheels. To combine the best of both worlds, this paper introduces the dynamic wheeled control including wheeled locomotion and in-situ wheel-to-foot (WtF) transformation of a full-sized wheel-biped transformable robot *SR600-II*. It can traverse on flat surfaces by wheels and transform to footed stance through its switching modules when facing obstacles. For wheeled locomotion, the kinematics considering upper-body lumped center-of-mass (CoM) constraint is first derived. Then, the dynamics of wheeled locomotion is modeled as a wheeled inverted pendulum (WIP) with variables related to the pose of upper body. After that, a parameter-varying linear quadratic regulator (LQR) controller is utilized to enable dynamic wheeled locomotion. For WtF transformation, the WtF balance constraints are first revealed. Then, a WtF transformation strategy is proposed to tackle the problem when robot transforms from wheeled balance state to in-situ biped stance state. It enables the robot to pass by the transition stages in which both wheels and feet touch the ground and to maintain its balance at the same time. Simulations and experiments on the *SR600-II* prototype have validated the efficacy of proposed dynamic wheeled control strategies for both wheeled locomotion and in-situ WtF transformation.

1. Introduction

As typical biomimetic robots, the research of biped robots has obtained fruitful results in design, modeling, and control [1]. Representative biped robots such as *ASIMO* [2], *Atlas* [3], *Digit* [4] show strong adaptability in human settlements. However, the powertrain of these highly capable biped robots is extremely inefficient compared with wheeled robots' [5]. Wheel-biped robots promise multiple locomotion modes, versatile environment adaptability with higher efficiency. They are able to pass through flat surface by wheels fast and efficiently, and negotiate obstacles by adjusting its upper-body pose or transform to biped walking. These characteristics make wheel-biped robots suitable for a wide range of living environment applications, such as inspection and logistics distribution.

In general, current wheel-biped robots can be classified into two types: wheel-biped transformable and non-transformable robots. For wheel-biped non-transformable robots, driving wheels, instead of robot feet, are installed at the end of robot shanks, and they move by wheels only. *Boston Dynamics* developed an electro-hydraulic wheel-biped robot *Handle* in 2017, it demonstrates highly agile mobility, such as robust two-wheeled balancing, jumping over obstacles and adapting to uneven terrain [6]. Inspired by *Handle*, *WLR* [7] and *WLR-II* [8] are also compact hydraulic wheel-biped robots and show a series of experiments including locomotion on rough pavement and carrying payload while squatting. A decoupling control framework consists of wheel balance controller and centroidal adjustment controller for *WLR-II* is proposed [9]. Klemm et al. presented a wheel-biped robot *Ascento* [10], it adopts a specific topology optimized four-bar-linkage leg mechanism to keep the CoM trajectory perpendicular to the ground when robot bending its legs. They modeled the full-body dynamics of *Ascento* and applied hierarchical quadratic optimization to realize whole-body control by including a LQR-assisted balancing task and other motion tasks [11]. Xin et al. [12] introduced a planning and control framework to realize online dynamic hybrid locomotion for wheel-biped robots. Chen et al. [13] studied the under-actuated motion planning and control of jumping for wheeled-biped robots. These wheel-biped non-transformable robots mentioned above show high agility in contrast with conventional bipedal robots. It is highly expected that if wheel-footed transformations and even biped walking can be achieved, the motion ability of these robots will be further enhanced.

For wheel-biped transformable robots, they have both wheel driving and bipedal walking modes, and can transform between them. Hashimoto et al. [14] presented a biped leg-wheeled robot *WL-16* in 2005, which is able to walk or move by wheel according to the ground conditions. Each leg of *WL-16* is formed by a 6 degrees of

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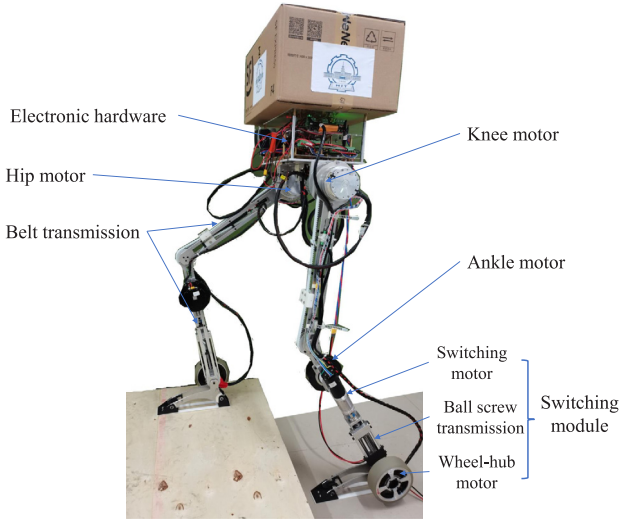

Fig. 1. The proposed wheel-biped transformable robot *SR600-II*.

freedom (DoFs) parallel mechanism with a wheel-driven foot. However, the structure of *WL-16* is complicated because of its parallel mechanism, and its four-wheeled locomotion method has poor agility. *DRC-HUBO+* [15] is a famous wheeled humanoid robot which can also transform between biped walking and wheeled driving. When transforming from biped walking to wheeled driving, it has to adjust its pose from standing state to kneeling posture with dramatic configuration change and vise versa. Besides, when in wheeled mode, the lower body of robot is equivalent to a fixed four-wheel chassis, which is hard to adjust robot height.

In our previous work, a 6-DoFs wheel-biped robot *SR600* was developed and achieved its dynamic wheeled balance using proportional integral differential (PID) approach [16,17]. In order to achieve wheel-footed transformation, we further built a wheel-biped transformable robot *SR600-II* as shown in Fig. 1. In this paper, we mainly focus on the dynamic wheeled motion control of such robots including wheeled locomotion and in-situ wheel-to-foot (WtF) transformation, which are representative motions that can truly combine the advantages of wheels and biped robots. Because of the coupling of wheels and robot upper-body dynamics in wheeled mode as well as the critical transition stages in WtF transformation process, the two-wheeled locomotion and WtF transformation are challenging problems for wheel-biped robots.

The contributions of this paper are summarized as follows. First, a novel wheel-biped transformable robot *SR600-II* is proposed. It is a 10-DoFs robots with two-wheeled locomotion and planar walking abilities, and switching modules are designed to ensure smooth and fast mode transformation. Second, the kinematics and dynamics model considering robot upper-body motions are derived and a parameter-varying LQR controller is utilized to realize dynamic wheeled balance and locomotion. Based on the kinematics model, we can keep the upper-body CoM moving along a straight line smoothly by proper joint trajectory generation method. As the motion of its upper body, we model its wheeled dynamics as a WIP with varying pendulum length and inertia according to different upper-body pose. Then, a parameter-varying LQR state feedback controller is applied. Third, we proposed a in-situ WtF transformation strategy to tackle the problem when robot transforms from wheeled balance state to in-situ bipedal stance state. In our method, we divide the transformation process into 3 stages: wheeled balance, WtF critical transition, and footed standing. The method enables robot to retract wheels by switching modules, pass by critical transition stage in which both wheels and feet touch the ground, and then transform to biped stance.

In the rest of this paper, we first describe the mechatronic design and control architecture of the wheel-biped robot in Section 2. Next, the modeling of the robot including kinematics, dynamics and WtF balance constraints in Section 3 is introduced. Then, dynamic wheeled control strategies used for wheeled locomotion and WtF transformation are presented in Section 4. In Section 5, simulations and experiments are conducted to validate the efficacy of proposed modeling and control strategies. Finally, we draw conclusions and discuss about the future work in Section 6.

2. Robot description

In order to realize wheel-footed transformation, we developed a wheel-biped transformable robot *SR600-II* based on our previous *SR600* robot. It has only lower limbs and waist with reference to the dimensions of a 175 cm human [18] designed for human settlement applications. On the flat ground, *SR600-II* can move by two-wheeled self-balancing approach and avoid vertical obstacles or complete specific tasks by adjusting it upper-body poses. When facing challenging terrains such as stairs and ditches, it can transform from wheeled balance state to bipedal standing at a ready state to walk. The switching module is designed to ensure the mode transformation, and a control architecture is introduced for the robot system.

2.1. Mechatronic design

The proposed system *SR600-II* is a 10-DoFs adult-sized wheel-biped transformation robot. It can be regarded as the combination of a planar biped walker with feet and two switching modules with driving wheels, as shown in Fig. 1. Each leg of the planar walker has 3-DoFs (the hip joint, knee joint and ankle joint each has a DoF actuated by servo motor), and the rotation axes of joints are parallel to each other. The legs are attached to the waist which houses most of the robot electronics like Inertial Measurement Unit (IMU), battery, motor drivers and controller boards. In order to reduce the inertia of robot legs, the knee joint motors are placed closed to the hip joints and the ankle joint motors to the knee joints, and the motor powers are transmitted through synchronous belt.

The transformation module is composed of a servo motor, a ball screw transmission mechanism and a customized wheel-hub motor. When the motor drives the ball screw to rotate, the wheel-hub motor translates up and down as it is attached to the slider of the ball screw. In this way, the mode transformation between wheeled driving and footed walking can be achieved. As for the wheel-hub motor, it is developed with 6 Nm of rated torque and only 735 g in weight, and an incremental encoder is installed on it.

To fulfill the system's performance requirements, the electronics of our robot system is carefully chosen. In the selection of hip, knee and ankle joint actuators, we use the *SolidWorks Motion* software to simulate the torque and velocity of these joints. According to the simulation results, we choose the integrated servo motor *INNFOS QDD-6010-64* as the hip and knee joint actuator. It is a powerful motor with 500 W rated power and rated torque up to 40 Nm, yet weights only 720 g. In order to perceive the critical transition state when wheels and feet touch the ground simultaneously, the quasi direct drive motor *T-Motor AK80-9* is chosen for the ankle actuation and foot contact sensing. The motor adopts a single-stage planetary gear reducer with a reduction ratio of 9, which ensures the transparency of force transmission. When robot reaches the transition stage, the current ($I_{ank} = K \cdot \tau$) of the ankle motor will increase sharply. In this way, the critical transition stage of wheel-biped transformation can be sensed by current value of ankle motor instead of mounting 6-dimensional force sensors on foot. *T-Motor AK80-6* motors are utilized in the wheel-hub motor, and *TLE5012B* encoders is installed for precise position and velocity feedback. For torque control of the wheel-hub motor, an *ODrive* is used for two *T-Motor AK80-6* motors. Two *DJI RM35* motors with *RMDS-108* drivers are selected to actuate the ball screw of the switching modules. All ten motors communicate via a control area network (CAN). To get robot

Table 1
Components and Suppliers of *SR600-II*.

Component	Name
Hip and kneel actuator	INNFOS QDD-6010-64
Ankle actuator	T-Motor AK80-9
Wheel motor	T-Motor AK80-6
Wheel motor encoder	TLE5012B
Wheel motor driver	ODrive Version 3.6
Switching module motor	DJI RM35
Switching module driver	RoboModule RMDS-108
IMU	Wit WT61PC (200 Hz)
Onboard controller	ARM-Cortex STM32F407

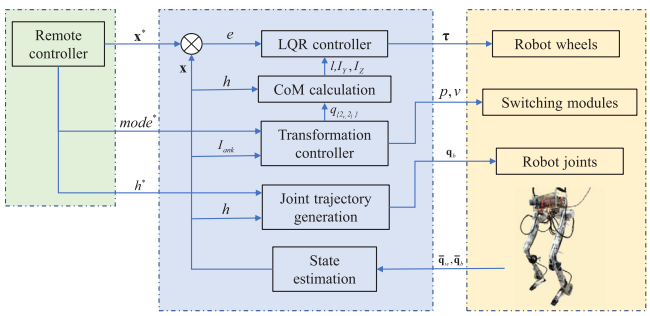

Fig. 2. Diagram of the control architecture for *SR600-II*. The left part in green is the remote controller that outputs target robot height h, desired motion mode and locomotion state variable. The middle part in blue is the main controller and the right part in yellow denotes the robot hardware.

pose, an IMU (*Wit WT61PC*) with maximum 200 Hz data transmission frequency is mounted at the robot waist. At last, we choose an ARM-Cortex STM32F4 microcontroller as the lower computing unit and it communicates with a PC by Bluetooth. A complete component list is shown in Table 1.

2.2. Control architecture

In order to realize effectively real-time control and computationally efficiency for the robot system, a control architecture is designed as shown in Fig. 2.

The user input originates from the remote PC and transfers to *SR600-II*, and the robot sends its state information to PC at the same time through Bluetooth devices bilaterally. The main controller running on robot mainly includes the parameter-varying LQR controller for wheeled locomotion, transformation controller for WtF mode transforming, CoM calculation to update robot physical parameters, joint trajectory generation and state estimation. The state estimation block gets robot joint angles, IMU data, wheeled encoder information, current of the ankle joint motor. And it outputs self-defined state variable, robot waist height calculated by forward kinematic model, mode switching signal and so on. The tasks of CoM calculation block are computing the upper-body lumped CoM position and inertia for LQR controller to update the robot parameters. The data processing of sensors, CAN communication for motors, control algorithm implementations and et al. are all carried out in an ARM-Cortex STM32F4 running μC/OS-III real-time operation system using C/C++.

3. Modeling

To apply the model-based dynamic wheeled control including wheeled locomotion and in-situ WtF transformation in Section 4, the system model to describe the robot *SR600-II* is derived. It includes the kinematics considering upper-body lumped CoM constraint, the dynamics for wheeled locomotion related to the robot pose and W2F balance constraints for mode switching. This paper mainly focuses on

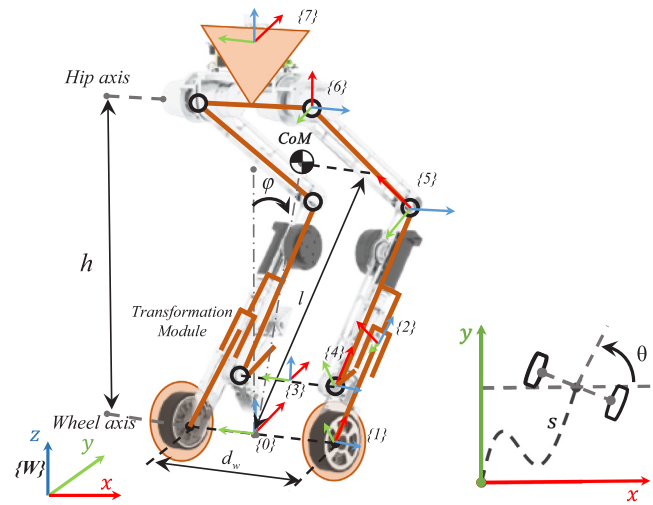

Fig. 3. Generalized coordinates and physical parameters of *SR600-II* in diagonal and top view.

the dynamic wheeled locomotion considering the upper-body dynamics and the in-situ transformation problems, therefore, identical motions are considered on right and left legs.

3.1. Coordinates and assumptions

Before systematically modeling the system, the coordinates, robot physical parameters and a set of state variables are introduced as shown in Fig. 3.

Frame {0} is attached at the midpoint of line segment connecting two driving wheels, and frame {3} the midpoint of that connecting two ankle joints. Frame {7} is set at the waist where IMU mounted. $q_{\{i_r,i_l\}}$ and $\tau_{\{i_r,i_l\}}$ denote the corresponding joint angle and torque on the ith joint on right (r) and left (l) sides, respectively. As right and left legs have identical motions and robot moves in sagittal plane, variables on the right side is used only for brevity hereon. φ denotes the forward tilt angle of upper-body lumped CoM, and l the distance from the origin of frame {0} to CoM. The upper body of the robot refers to the part above the wheels in wheeled locomotion mode, and when in footed standing mode, it refers to the part of the robot above the feet. d_w is the distance between two wheels, and h the distance from the hip axis to the wheel axis. For odometry, we use x and y to represent the robot position in world frame {W} and s as the traveled distance. θ is the steering angle.

For our robot, we choose generalized coordinates as

$$\mathbf{q} = \begin{bmatrix}\mathbf{q}_b & \mathbf{q}_w\end{bmatrix}^{\mathrm{T}}, \tag{1}$$

where

$$\mathbf{q}_b = \begin{bmatrix}q_{1_r} & q_{2_r} & q_{4_r} & q_{5_r} & q_{6_r}\end{bmatrix}^{\mathrm{T}} \tag{2}$$

indicates the robot upper-body generalized coordinates of joints, and $\mathbf{q}_w$ denotes the generalized coordinates for wheeled locomotion. In addition, q_{1_r} denotes the rotation angle of shank, q_{2_r} the link offset of the switching module, and q_{4_r}, q_{5_r}, q_{6_r} are rotation angle of ankle, knee and hip joint, respectively.

To simplify the model, we draw the following assumptions: (1) The joints are assumed with no friction either the hysteresis exist; (2) The no-slip condition are assumed to simplify the friction between floor and wheels; (3) All links and bodies are considered as rigid; (4) The joint motor dynamic and the system delay are neglected.

3.2. Joint kinematics considering CoM constraint

In this subsection, we first establish the kinematics model of the robot joints by Denavit–Hartenberg (D–H) method. Then, the upper-body lumped CoM position and the tilt angle φ is calculated. After that, the joint kinematics considering upper-body lumped CoM constraints and the mapping from robot waist height h to joint variables $\mathbf{q}_b$ are derived to guide the planning of joint trajectory and the control of robot pose.

To achieve the dynamic wheeled and in-situ footed balance, the wheel-biped system should meet the CoM constraint. For wheeled locomotion, it means the lumped CoM of upper body should be placed right above the perpendicular plane of wheel axis. And for static footed balance, the upper-body lumped CoM needs to fall within the support polygon formed by feet.

According to the D–H method, the transform matrix of adjacent links is

$$ {}^{i-1}_{i}\mathbf{T} = \begin{bmatrix} {}^{i-1}_{i}\mathbf{Q} & {}^{i-1}\mathbf{p}_{Oi} \\ 0 & 1 \end{bmatrix}, \tag{3} $$

where ${}^{i-1}_{i}\mathbf{Q}$ and ${}^{i-1}\mathbf{p}_{Oi}$ are rotation matrix and translation vector, respectively. The transform matrix of each link is obtained by Eq. (3). Further, we can get the transform matrix from waist frame $\{7\}$ to frame $\{0\}$ as ${}^{0}_{7}\mathbf{T} = {}^{0}_{1}\mathbf{T} \cdot {}^{1}_{2}\mathbf{T} \cdot {}^{2}_{5}\mathbf{T} \cdot {}^{5}_{6}\mathbf{T} \cdot {}^{6}_{7}\mathbf{T}$ for wheeled driving, and waist frame $\{7\}$ to ankle frame $\{3\}$ as ${}^{3}_{7}\mathbf{T} = {}^{3}_{4}\mathbf{T} \cdot {}^{4}_{5}\mathbf{T} \cdot {}^{5}_{6}\mathbf{T} \cdot {}^{6}_{7}\mathbf{T}$ for biped stance.

According to the general formula to calculate CoM of a multi-body system, the upper-body CoM position with respect to frame $\{0\}$ can be obtained as

$$ {}^{0}\mathbf{p}_{CoM} = \frac{\sum \left({}^{0}\mathbf{p}_{Oi} + {}^{0}_{i}\mathbf{R} \cdot {}^{i}\mathbf{p}_{Ci}\right) m_i}{\sum m_i}, \tag{4} $$

where ${}^{0}\mathbf{p}_{Oi}$ and ${}^{0}_{i}\mathbf{R}$ are the translation vector from the origin of frame $\{i\}$ to frame $\{0\}$ and rotation matrix from frame $\{i\}$ to frame $\{0\}$, respectively. Besides, m_i denotes the mass of ith link, and ${}^{i}\mathbf{p}_{Ci}$ the CoM position of ith link expressed in its local frame $\{i\}$.

From Eq. (4), the upper-body lumped CoM position ${}^{0}\mathbf{p}_{CoM} = \left[{}^{0}p_{CoM_x}, {}^{0}p_{CoM_y}, {}^{0}p_{CoM_z}\right]^{\mathrm{T}}$ is a vector with respect to robot current pose $\mathbf{q}_b$, and we can further get the tilt angle of the CoM as

$$ \varphi = \arctan\left(\frac{{}^{0}p_{CoM_x}}{{}^{0}p_{CoM_z}}\right). \tag{5} $$

To achieve the wheeled balance, the tilt angle needs to meet $\varphi = 0$, and we constrain the waist frame $\{7\}$ aligned to frame $\{0\}$, which means the waist of robot will keep horizontal. Combining with the relationship between h to joint variables and the CoM constraint, we can formulate the joint kinematics considering upper-body CoM constraints as

$$ \begin{cases} h = \left(L_{sw} + q_{2_r}\right) \sin q_{1_r} + L_t \cos q_{5_r} \\ {}^{0}p_{CoM_x} = 0 \\ q_{1_r} + q_{5_r} + q_{6_r} = \pi \end{cases}, \tag{6} $$

where L_{sw} denotes the initial shank length of the branch where the switching module is located, and L_t the length of thigh. It is noted that Eq. (6) has multiple solutions. It can be handled by making the translation of robot switching module $q_{\{2_r,2_l\}} = 0$ in wheeled locomotion, and move by a giving ${}^{ref}q_{\{2_r,2_l\}}$ in W2F transformation.

By solving (6), the process of calculating the inverse kinematics of upper body with CoM constraint, we can get the relationship between h and joint variables $\mathbf{q}_b$. To simplify, it can be written as

$$ \mathbf{q}_b = {}^{\mathbf{q}_b}\mathbf{T}_h \cdot h, \tag{7} $$

where ${}^{\mathbf{q}_b}\mathbf{T}_h$ is the mapping from h to each joint variable $\mathbf{q}_b$. From (7), when giving a desired robot height h, we can get the corresponding joint rotations and translations satisfying the CoM constraint according to it.

3.3. Wheeled motion and dynamics

As shown in Fig. 3, for two-wheeled robot, it satisfies non-holonomic constraints and WIP model is extensively used to simplify its dynamics. Meanwhile, many linear and non-linear control methods are successful applied to achieve its stabilization and locomotion [19]. However, the classical WIP model cannot model the dynamics of wheel-biped robots accurately due to the upper-body motion varying CoM position and moment of inertia.

(1) Wheeled Kinematics: For wheeled locomotion of our wheel-biped robot, we introduce tilt angle φ, traveling distance s and steering angle θ to describe the two-wheeled motion. So, the generalized coordinates is chosen as

$$ \mathbf{q}_w = [\varphi \quad s \quad \theta]^{\mathrm{T}}. \tag{8} $$

We indicate n_u the dimensions of $\mathbf{q}_w$. By differentiating Eq. (8), we can get

$$ \dot{\mathbf{q}}_w = [\dot{\varphi} \quad v \quad \omega]^{\mathrm{T}}, \tag{9} $$

where $v = \dot{s}$ denotes the linear and $\omega = \dot{\theta}$ the angular velocity of the robot.

Furthermore, the linear and angular velocity of robot can be represented by angular velocity of right and left wheels as

$$ \begin{cases} v = \frac{r}{2}\left(\dot{\theta}_r + \dot{\theta}_l\right) \\ \omega = \frac{r}{d}\left(\dot{\theta}_r - \dot{\theta}_l\right) \end{cases}, \tag{10} $$

where r stands for the wheel radius, θ_r and θ_l denote the rotation angles of right and left wheels, respectively.

(2) Wheeled Dynamics: As the motion of the upper body introduces changes of robot CoM position and moment of inertia, we model the wheeled dynamics of the wheel-biped robot as a WIP with variables related to its upper-body poses.

To formulate the equation of robot wheeled dynamics, the Lagrange method is applied. First, the Lagrange energy function $L = K - V$ should be obtained. Here, K and V are the total kinetic and potential energy of the robot system, respectively. Then, according to the Lagrange equations of the first kind

$$ \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\right) - \frac{\partial L}{\partial q} = J^{\mathrm{T}}\tau, \tag{11} $$

we can formulate its dynamics equation as

$$ \mathbf{M}\left(\mathbf{q}_w\right)\ddot{\mathbf{q}}_w + \mathbf{V}\left(\mathbf{q}_w, \dot{\mathbf{q}}_w\right) + \mathbf{G}\left(\mathbf{q}_w\right) = \mathbf{J}^{\mathbf{T}}\boldsymbol{\tau}. \tag{12} $$

Here, $\mathbf{M}\left(\mathbf{q}_w\right) \in \mathbb{R}^{n_u \times n_u}$ donates the symmetric mass matrix, $\mathbf{V}\left(\mathbf{q}_w, \dot{\mathbf{q}}_w\right) \in \mathbb{R}^{n_u \times n_u}$ donates the vector of Coriolis and centrifugal terms, and $\mathbf{G}\left(\mathbf{q}_w\right) \in \mathbb{R}^{n_u}$ is the vector of gravity terms. The matrix $\mathbf{J}\left(\mathbf{q}_w\right) \in \mathbb{R}^{n_u \times n_u}$ selects the torques $\boldsymbol{\tau}$ added to driving wheels.

Finally, by taking into robot physical parameters, the wheeled dynamics equation considering the motion of upper body is

$$ \begin{bmatrix} M_b l^2 + I_Y & M_b l \cos\varphi & 0 \\ M_b l \cos\varphi & M_b + 2M_w + \frac{2I_w}{r^2} & 0 \\ 0 & 0 & \frac{d^2}{2}M_w + \frac{d^2}{2r^2}I_w + I_Z \end{bmatrix} \begin{bmatrix} \ddot{\varphi} \\ \dot{v} \\ \ddot{\theta} \end{bmatrix} + \begin{bmatrix} -M_b g l \sin\varphi \\ -M_b l \sin\varphi \dot{\varphi}^2 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ \frac{1}{r} & \frac{1}{r} \\ \frac{d}{2r} & -\frac{d}{2r} \end{bmatrix} \begin{bmatrix} \tau_r \\ \tau_l \end{bmatrix}, \tag{13} $$

where M_b denotes the upper-body lumped mass and M_w the mass of wheel, I_w denotes the inertia of wheel about its rotational axis, I_Y is the moment inertia about the wheel axis and I_Z the moment of inertia about the yaw angle.

For the WIP-based parameter-varying dynamic model, l, I_Y and I_Z are variables related to the upper-body pose $\mathbf{q}_b$. As $\mathbf{q}_b$ can be fully represented by the waist height h, we can formulate the relationship between these variables and h.

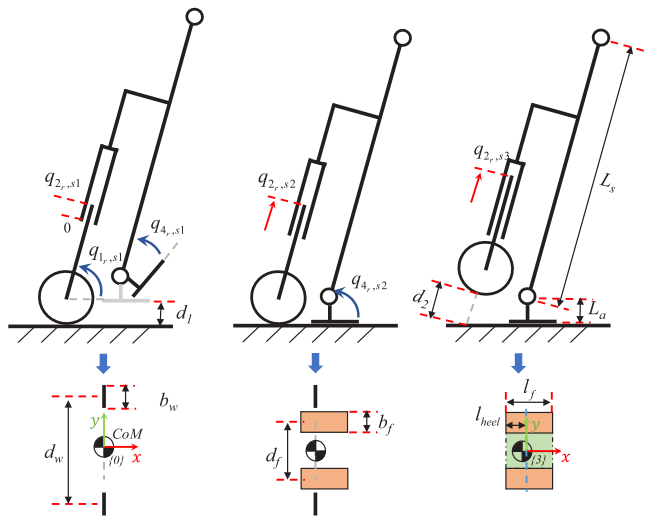

Fig. 4. Three stages of in-situ WtF transformation process in side view of shanks, and the regions of wheels and feet in contact with the ground. (a) The initial wheeled balance stage. (b) The critical transition stage in which wheels and feet touch the ground simultaneously. (c) The footed standing stage.

When giving a certain h, we can calculate the distance from the lumped CoM to the origin of frame $\{0\}$ as

$$l(h) = \left\| {}^{0}\mathbf{p}_{CoM} \right\| , \tag{14}$$

which represents the length of pendulum. In addition, I_Y and I_Z can be determined as

$$I_{\{Y,Z\}}(h) = \sum m_i \cdot {}^{\{Y,Z\}}r_{m_i}^2 , \tag{15}$$

where ${}^{\{Y,Z\}}r_{m_i}^2$ indicates the distance from the CoM of ith link to the Y or Z axis of frame $\{0\}$.

3.4. WtF balance constraints

In order to realize the in-situ WtF transformation stably, the balance constraints for wheel-biped transformation are revealed. As shown in Fig. 4, the transformation process can be divided into three stages — the wheeled balance stage, the critical transition stage and the footed standing stage.

In the wheeled balance stage as shown in Fig. 4(a), the robot is natural unstable because of the linear contact with the ground. To keep wheeled stabilization, the vertical projection of lumped upper-body CoM needs to fall on the wheel axis, ideally. In sagittal and coronal plane, the x and y coordinates of CoM position should meets

$$\begin{cases} {}^{0}p_{CoM_x} = 0 \\ {}^{0}p_{CoM_y} \le \left| \frac{d_w + b_w}{2} \right| \end{cases} , \tag{16}$$

where b_w is the width of wheel.

When the switching module retracts before robot feet touch the ground, the system is still in the wheeled balance state. Hence, the wheeled dynamics model derived in Section 3.3 can maintain wheeled balance by adjusting its upper-body poses.

At the critical transition stage as shown in Fig. 4(b), both wheels and feet touch the ground. It is a complicated transient state involving motion of wheels and feet, as well as transforming from linear contact to plane contact with the ground. The translation distance of the switching module from initial wheeled balance state to the critical transition state is

$$d_1 = r - L_a + \left(L_{sw} + q_{2_r,s1} - L_s \right) \sin q_{1_r,s1} , \tag{17}$$

where $q_{2_r,s1}$ and $q_{1_r,s1}$ denote value of joint variables q_{2_r} and q_{1_r} at stage 1. L_a and L_s represent the length from ankle to foot and the length of shank, respectively. Meanwhile, the ankle joint rotates from initial $q_{4_r,s1}$ to $q_{4_r,s2}$ and contact the ground stably. As the foot is light in wight, it is regarded as $q_{4_r,s2} \approx q_{1_r,s1}$.

As for footed stance state as shown in Fig. 4(c), the robot maintains balance by its feet and the switching module retracts to the minimum length. The vertical projection of upper-body CoM should fall within the support polygon formed by feet. The footed upper-body CoM position ${}^{3}\mathbf{p}_{CoM} = \left[\begin{array}{ccc} {}^{3}p_{CoMx} & {}^{3}p_{CoMy} & {}^{3}p_{CoMz} \end{array} \right]^{\mathrm{T}}$ can be calculated the same as (4), so its x and y coordinates needs to satisfy

$$\begin{cases} -l_{heel} \le {}^{3}p_{CoMx} \le \left(l_f - l_{heel} \right) \\ {}^{3}p_{CoMy} \le \left| \frac{d_f + b_f}{2} \right| \end{cases} , \tag{18}$$

where l_f is the length of robot foot, and l_{heel} is the distance from heel to the rotation axis of foot.

4. Dynamic wheeled control

Based on the model derived in Section 3, the dynamic wheeled motion control for wheel-biped transformable robots is introduced in this section. First, the wheeled locomotion control is decoupled into joint trajectory generation of upper body and the WIP-based wheeled locomotion. Then, we propose an in-situ WtF transformation strategy to tackle the mode switching problem.

4.1. Wheeled locomotion control

(1) Joint Trajectory Generation: From the kinematic model derived in Section 3.2, the lumped CoM trajectory of upper body should move along a straight line when robot changing its altitude. Therefore, the joint trajectories needs to be properly generated to satisfy the CoM constraint.

In Eq. (7), the relationship between h and joint variables is revealed. By differentiating it, we can further get

$$\dot{\mathbf{q}}_b = {}^{\mathbf{q}_b}\mathbf{J}_h \cdot \dot{h} , \tag{19}$$

where ${}^{\mathbf{q}_b}\mathbf{T}_h$ is Jacobian matrix mapping from $\dot{h}$ to the joints velocity.

In Fig. 2, a joint trajectory generation block is utilized to control the movement of robot joints to reach the desired height ${}^{*}h$ which is one of the control variable. The block calculates the bias between desired ${}^{*}h$ and the robot current h, then it can generate the $\mathbf{q}_b$ and $\dot{\mathbf{q}}_b$ of body joints according to Eqs. (7) and (19), and finally transmits the sampling joint trajectory instructions to corresponding actuators.

(2) WIP-based Wheeled Locomotion: For wheel-biped robots, due to the motion of robot upper body, the CoM position and the moment of inertia of upper body will keep changing. So, a robust wheeled locomotion controller is of great significance for the robot. LQR is an optimal state feedback control method to regulate linear systems at the minimum cost, which is successfully used in controlling WIP systems and shows high reliability and robustness in wheeled balance control.

According to the kinematics and dynamics model derived in Section 3.3, we choose the state variable of the system as

$$\mathbf{x} = \left[\mathbf{q}_w, \dot{\mathbf{q}}_w \right]^{\mathrm{T}} . \tag{20}$$

By linearizing the WIP-based non-linear dynamics model about the equilibrium point $\left(\mathbf{x}_0, \mathbf{u}_0 \right) = (\mathbf{0}, \mathbf{0})$, we can obtain the linear state-space equation of our robot as

$$\dot{\mathbf{x}} = \mathbf{A}(h)\,\mathbf{x} + \mathbf{B}(h)\,\mathbf{u} , \tag{21}$$

where $\mathbf{A}(h)$ and $\mathbf{B}(h)$ are the state and input matrices respectively, and they both contains variables $l(h)$, $I_Y(h)$ and $I_Z(h)$. The system input $\mathbf{u} = \left[\tau_r, \tau_l \right]^{\mathrm{T}}$ indicates torque applied to driving wheels. According to the Lyapunov first stability criterion, the system is inherent unstable, yet controllable and observable.

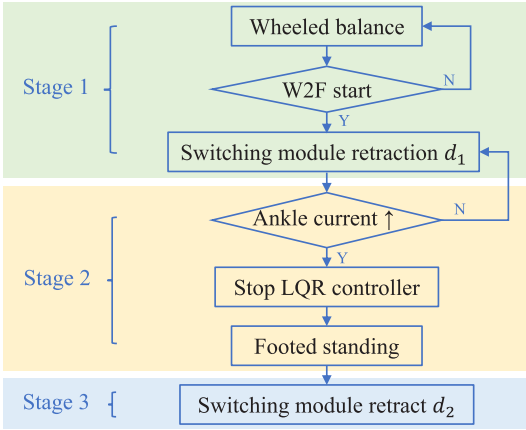

Fig. 5. The in-situ WtF transformation control process. Stage 1: wheeled balance. Stage 2: WtF critical transition. Stage 3: Footed standing.

As the $\mathbf{A}(h)$ and $\mathbf{B}(h)$ have variables, so a parameter-varying LQR controller is designed to update the parameters as well as track the desired state $\mathbf{x}^*$ giving by users.

We first form the feedback control law as

$$\mathbf{u} = -\mathbf{K}\mathbf{e}, \tag{22}$$

where the state error $\mathbf{e} = \mathbf{x}^* - \mathbf{x}$. Then, the cost function of the LQR controller can be formulated as

$$J = \int_{t_0}^{t_f} \left(\mathbf{e}^{\mathrm{T}}\mathbf{Q}\mathbf{e} + \mathbf{u}^{\mathrm{T}}\mathbf{R}\mathbf{u}\right) dt, \tag{23}$$

where $\mathbf{Q}$ and $\mathbf{R}$ are the state and control weight matrix defined by users. In order to get the state feedback matrix $\mathbf{K}$, the Riccati equation is established as

$$\mathbf{A}^{\mathrm{T}}(h)\mathbf{P} + \mathbf{P}\mathbf{A}(h) - \mathbf{P}\mathbf{B}(h)\mathbf{R}^{-1}\mathbf{B}^{\mathrm{T}}(h)\mathbf{P} + \mathbf{Q} = \mathbf{0}. \tag{24}$$

By solving Eq. (24), we can get the matrix $\mathbf{P}$ and then the state feedback matrix can be obtained as

$$\mathbf{K} = \mathbf{R}^{-1}\mathbf{B}^{\mathrm{T}}\mathbf{P}. \tag{25}$$

As shown in Fig. 2, when giving desired control variables ${}^*h$ and ${}^*\mathbf{x}$ in the closed-loop of wheeled locomotion, h will first be calculated by the feedback of joint variables, and then $l(h)$ $I_Y(h)$ and $I_Z(h)$ will be updated by Eqs. (14) and (15). Finally, the LQR controller will update the states function and recalculate the feedback gain matrix $\mathbf{K}$ at each time step T_s.

4.2. WtF transformation strategy

The in-situ WtF transformation process can be divided into 3 stages: wheeled balance, WtF critical transition and footed stance. According to it, an in-situ WtF transformation strategy for wheel-biped robots is proposed, as shown in Fig. 5.

In the control process, at stage 1, the initial state of the robot is in-situ wheeled balance. When receiving the starting command giving by users through remote controller, the switching module begins to retract at a distance of d_1. Before robot feet touch the ground, robot is in wheeled balance state with varying shank length. So, the dynamic wheeled locomotion controller designed in Section 4.1 can be also applied to maintain the wheeled balance in WtF transformation process.

For the critical transition stage, robot first stops its wheeled balance controller and then transforms to the footed standing state. As derived in Section 3.4, the relation between wheeled and footed balance constraint regions can be formulated as

$$\begin{cases} \Omega\left({}^0p_{CoM_x}\right) \subset \Omega\left({}^3p_{CoM_x}\right) \\ \Omega\left({}^0p_{CoM_y}\right) \supset \Omega\left({}^3p_{CoM_y}\right) \end{cases}, \tag{26}$$

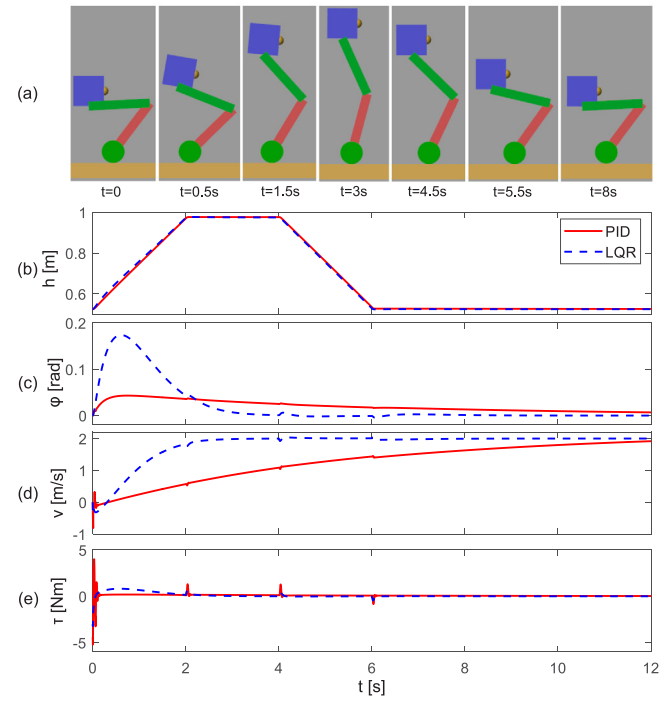

Fig. 6. The simulation results of the robot moving forward while varying its upper-body pose. (a) Snaps of the simulation using parameter-varying LQR approach; (b) Height of the waist; (c) Incline angle of the upper-body CoM; (d) Velocity of the robot; (e) Torque of wheel.

which indicates the footed standing is more stable than wheeled balance in sagittal plane, while wheeled stabilization better than footed standing in coronal plane. Therefore, at stage 2, as long as the CoM projection falls within the support polygon formed by feet, the WtF transformation can be executed directly.

In order to determine the critical transition state accurately, the current of ankle joint motors are measured during the whole wheeled balance stage. Once the robot feet touch the ground, their values will sharply increase, which indicates the critical transition stage is achieved.

Finally, at stage 3, the switching module continues retracting at a distance of d_2 and complete the WtF mode transformation.

5. Experiments & results

5.1. Simulation results of wheeled locomotion

In this part, we demonstrate the wheeled locomotion performance in simulation. A simplified virtual robot were constructed in *Simscape Multibody* with *ODE* solver and keep its physical parameters similar to the real prototype *SR600-II*. Its total mass is 14 kg, the rated torques exerted to joints and wheels were limited to 40 N m and 6 N m, respectively. The time step of the control loop is set as $T_s = 5$ ms.

(1) Dynamic Wheeled Locomotion: As shown in Fig. 6, the simulation demonstrates that the robot accelerates from the initial static state to desired velocity $v = 2$ m/s, and changes its waist height h from 0.5 m to 0.95 m simultaneously. For the parameter-varying LQR controller, we choose weight matrices $\mathbf{Q} = diag\,[10, 1, 1, 1, 1, 1]$ and $\mathbf{R} = diag\,[0.1, 0.1]$, and the feedback gain matrix $\mathbf{K}$ is updated according to Eqs. (24) and (25). Giving the desired state as $\mathbf{x}^* = [0, 0, 0, 0, 2, 0]^{\mathrm{T}}$, the simulation is launched. For comparison, the PID approach in our previous work [17] is used to handle the same task.

The simulation results show that the joint trajectory generation method can effectively convert the robot waist height h from 0.5 m to 0.95 m in Fig. 6(b), and maintain its balance by the both PID and

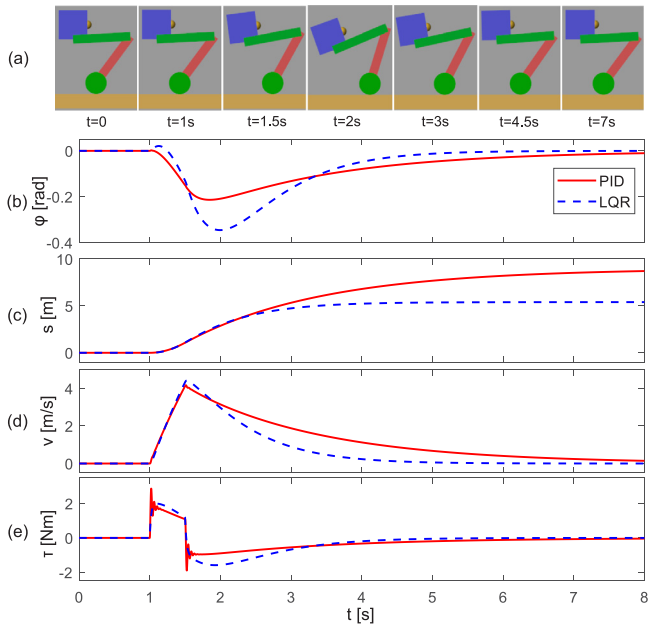

Fig. 7. The simulation results of the robot subjected to the external impact force (80 N). (a) Snaps of the simulation; (b) Incline angle fo the CoM; (c) Moving distance of the robot; (c) Velocity of the robot; (d) Torque of wheel.

parameter-varying LQR approaches. However, the parameter-varying LQR controller outperforms the PID approach by giving a shorter setting time and smoother torque output of wheels.

(2) Impact Robustness of Wheeled Locomotion: Further, we test the performance when robot dealing with external impact force. As shown is Fig. 7, an impact force of 80 N is added to the waist of robot horizontally at$t = 1$ s and lasting 10 ms. For the parameter-varying LQR controller, the weight matrices are set as $\mathbf{Q} = diag\,[100, 1, 1, 10, 1, 1]$ and $\mathbf{R} = diag\,[1, 1]$, and robot height h keeps at 0.5 m. Again, we use the PID approach as a comparison.

The simulation results indicate that, using the parameter-varying approach, the robot can sustain the impact force without falling and returns to equilibrium state in about 4 s, while the PID approach needs longer recovery time and farther moving distance. Furthermore, the robot can withstand a maximum impact force of 132 N using the parameter-varying controller, but the robot will fall down when the applied impact force exceeds 80 N using the PID approach.

(2) Rotational Motion: As shown in Fig. 8, the simulation demonstrates the robot rotation performance by rotating $\pi/3$ rad, and simultaneously changing the robot height from 0.5 m to 0.95 m. The weight matrices of the parameter-varying LQR controller are set the same as the simulation 1.

The simulation results show that the robot can rotate from the initial pose to the setting state smoothly.

5.2. Experimental results

(1) Dynamic Wheeled Locomotion: In this part, we demonstrate the wheeled stabilization when robot dynamically varying its upper body on *SR600-II* robot. As shown in Fig. 9, *SR600-II* first squats from the upright posture ($h = 0.95$ m) to $h = 0.75$ m, then drives forward about 1 m and finally extends to original upright pose. For the parameter-varying LQR controller, we choose $\mathbf{Q} = diag\,[10, 1, 0.1, 10, 1, 0.1]$ and $\mathbf{R} = diag\,[1, 1]$, and time step of the controller is 5 ms.

As we can see from Fig. 9, although the hardware of *SR600-II* is not specially optimized, the results show it can achieve the wheeled locomotion and dynamically changing its upper body based on the modeling and control approach presented before.

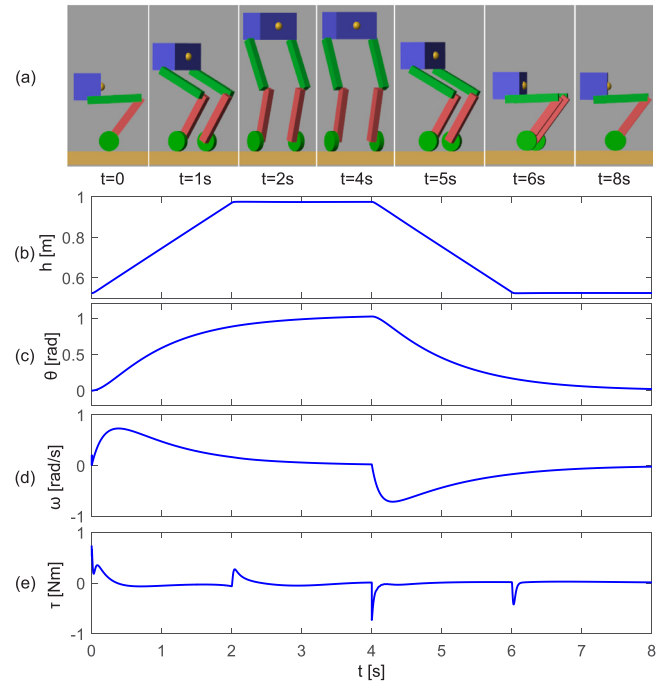

Fig. 8. The simulation results of the robot rotating while varying its upper-body. (a) Snaps of the simulation; (b) Height of the waist; (c) Rotation angle of the robot; (d) Angular velocity of the robot; (d) Torque of wheel.

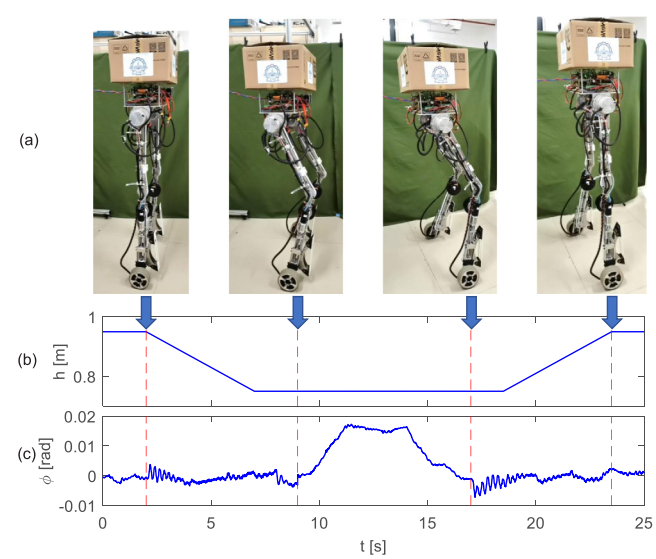

Fig. 9. *SR600-II* demonstrates dynamic wheeled locomotion while changing the height of upper body. (a) Snaps of the experiment; (b) Waist altitude of the robot; (c) Incline angle of the CoM.

(2) WtF In-situ Transformation: In order to validate the efficacy of proposed in-situ WtF control strategy, we demonstrate how *SR600-II* transforms from initially wheeled balance state to footed standing using its switching modules. As shown in Fig. 10, we set the robot height at $h = 0.75$ m in wheeled driving mode, and the initial length of translation variable $q_{2_{r,s1}} = 40$ mm. The robot is first in wheeled balance state in $0 \sim 2$ s. Then, the switching modules start to retract with velocity of $\dot{q}_{\{2_r,2_l\}} = 200$ mm/s at point P_1. At point P_2, *SR600-II* reaches the critical transition state, where both the wheels and feet touch the ground. Then, the LQR controller is turned off and the robot stars to stabilize by its feet while the switching modules continue retracting by $d_2 = 40$ mm.

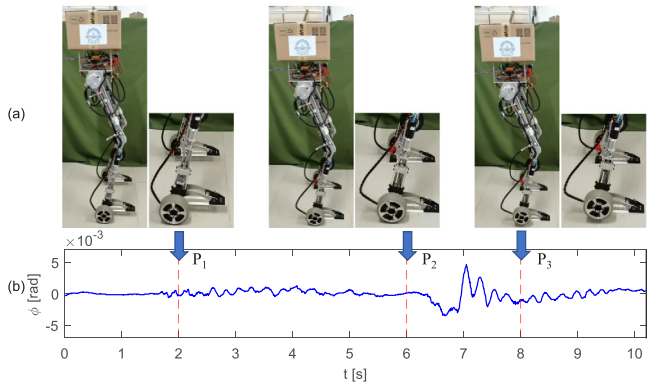

Fig. 10. *SR600-II* demonstrates in-situ WtF transformation. (a) Snaps of the experiment; (b) Incline angle of the CoM.

In Fig. 10(b), in wheeled locomotion mode (0~8 s), the average and maximum value of the robot tilt angle are 3.27×10^{-6} rad and 4.66×10^{-3} rad; and in footed stance state (8~10.2 s), the maximum backward and forward tilt angles are -1.38×10^{-3} rad and 0.84×10^{-3} rad, respectively. According to the balance constraints derived in Eqs. (16) and (18), the tilt angle should meet $\phi = 0$ when in wheeled balance state, and $-\arcsin\left(\frac{l_{heel}}{l(h)}\right) \leq \phi \leq \arcsin\left(\frac{l_f - l_{heel}}{l(h)}\right)$ that is $-0.093 \leq \phi \leq 0.347$ rad in footed standing state. The experimental result shows that the robot can successfully implement the in-situ WtF transformation satisfying the balance constraints by the strategy proposed in this paper.

6. Conclusions

This paper presented the dynamic wheeled motion control of a wheel-biped transformable robot *SR600-II* including the modeling and control of wheeled locomotion and in-situ WtF transformation. First, the kinematics model considering upper-body lumped CoM constraint and WIP-based wheeled dynamics model with variables were introduced. And we further derived the in-situ WtF balance constraints for mode switching. Then, joint trajectory generation method to keep the upper-body CoM move linearly, parameter-varying LQR controller for dynamic wheeled locomotion and control strategy to handle in-situ WtF transformation were verified by simulations and prototype experiments. In our future work, the dynamic transformation for both wheel-to-foot and foot-to-wheel will be investigated based on our robot. Moreover, the perception ability of the robot will be enhanced and practical applications of the robot will be achieved.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.birob.2021.100027.

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